

Chapter 15. Feedback Enhancements on Conventional Ventilator Breaths

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Conventional Positive-Pressure Breath-Delivery Strategies and Modes

The first generation of positive-pressure mechanical ventilators were simple high-pressure gas regulators on which clinicians could set the circuit pressure and the breathing frequency. In the middle of the twentieth century, more sophisticated devices appeared that allowed direct clinician control of flow and volume along with breath timing and expiratory pressure. As ventilator design improved and microprocessors became available, feedback mechanisms appeared that could provide automatic adjustments in these set variables depending upon a variety of conditions.¹ A simple example was the use of a patient-effort sensor to adjust the number of mechanical breaths provided during assist-control modes or synchronized intermittent mandatory ventilation.^{2–4} A variation on this breath rate feedback mechanism was mandatory (or minimum) minute ventilation, which used minute ventilation to adjust the number of positive-pressure breaths delivered.⁵ At the same time, the development of flow control valves that could be adjusted based on a clinician-selected airway-pressure target appeared.^{6–9} This gave clinicians the choice of using either set flow-volume-targeted modes (volume assist-control ventilation; volume-targeted synchronized intermittent mandatory ventilation [volume SIMV]) or pressure-targeted modes (pressure-targeted assist-control ventilation [PACV]; pressure support; pressure SIMV). Taken together, these flow-targeted, volume-targeted, and pressure-targeted strategies comprise what is commonly referred to today as “conventional” mechanical ventilation.

In the late twentieth and early twenty-first century, increasingly complex and clever feedback mechanisms for these conventional breaths have been introduced. The behavior of these features is made more understandable if one considers that all positive-pressure breaths can be described by three variables: the trigger variable (what initiates a breath—usually an effort sensor or a machine timer); the target variable (what governs gas delivery—usually either a set flow or a set pressure target); and the cycle variable (what terminates the breath—usually a set time, flow, or volume).^{6,8} These newer feedback systems are designed to adjust one or more of these breath-delivery variables based on prescribed algorithms to provide more physiologically targeted and patient interactive ventilatory support. In addition, there has also been the development of feedback mechanisms to adjust the fractional inspired oxygen concentration (FI_{O_2}) and positive end-expiratory pressure (PEEP). These feedback mechanisms are often considered “partial” closed-loop conventional mechanical ventilation. The clinician, however, cannot expect these features to provide safe and effective support automatically. Rather, the clinician must understand their rationale, design principles, lung-protective targets, and evidence-based outcomes to apply them properly. These are the focus of the remainder of this chapter.

Feedback Control of Combination Pressure-Targeted and Flow-Targeted Breaths

The advantage of a flow-targeted, volume-cycled breath is that a guaranteed volume is delivered with every breath (even though applied airway pressures may change). The advantages of a pressure-targeted breath (either time- or flow-cycled) are that an airway pressure limit is guaranteed (even though volumes may change), that the rapid initial flow may enhance gas mixing, and that the variable-flow pattern may enhance patient-ventilator synchrony.^{7,10,11}

Over the last two decades, a number of engineering innovations have attempted to combine these features by producing feedback algorithms that allow some control of volume with pressure targeting and some control of pressure with flow targeting. Collectively, these are often referred to as *hybrid breaths* or *dual-control breaths*.^{12–15} On the current generation of mechanical ventilators, there are two basic approaches to providing these types of breaths: dual control within a breath (DCWB; intrabreath control) and dual control from breath to breath (DCBB; interbreath control). The former uses a measured flow input to switch from pressure targeting to flow targeting in the middle of the breath. The latter uses a measured volume input to manipulate the pressure level of subsequent pressure-targeted breaths.

Dual Control Within a Breath

The currently available DCWB breath begins with either patient or machine triggering and is followed by a pressure-targeted flow-delivery algorithm.¹⁶ There is thus a high initial flow to rapidly pressurize the airway and then subsequent flow adjustments according to respiratory system mechanics and patient effort to maintain the target pressure. As the lungs fill, flow decelerates until a flow-cycling mechanism terminates the breath. In these respects, this breath type is similar to the pressure support breath. Unlike pressure support, however, the clinician also must set a minimum tidal volume, flow, and backup rate with the DCWB breath. These backup settings take over control of the breath should the pressure-targeted flow drop below the minimum required to deliver the set tidal volume in the allotted inspiratory time. The breath thus begins like pressure support and can either flow cycle like pressure support (if the volume meets or exceeds the set minimum) or volume cycle like a flow-targeted breath (if necessary to deliver the set volume).^{12–14,16}

The reasoning behind DCWB breaths is that the high initial flow provides better gas mixing and also reduces flow dyssynchrony during assisted breaths, whereas the volume guarantee ensures a constant tidal volume (V_T).^{10,17–21} DCWB breaths thus can be considered to be “more comfortable” flow-targeted, volume-cycled breaths. Alternatively, they could be considered pressure support with a V_T “safety net.”¹⁶

DCWB breath algorithms exist on several ventilators, but their clinical role remains unclear because the few studies looking at this strategy have generally focused only on device performance.^{16,22} Indeed, DCWB mode use is driven primarily by a clinician’s belief in the underlying concept. In recent years, DCWB modes have gradually disappeared and have been replaced with the simpler to use DCBB approach described below.

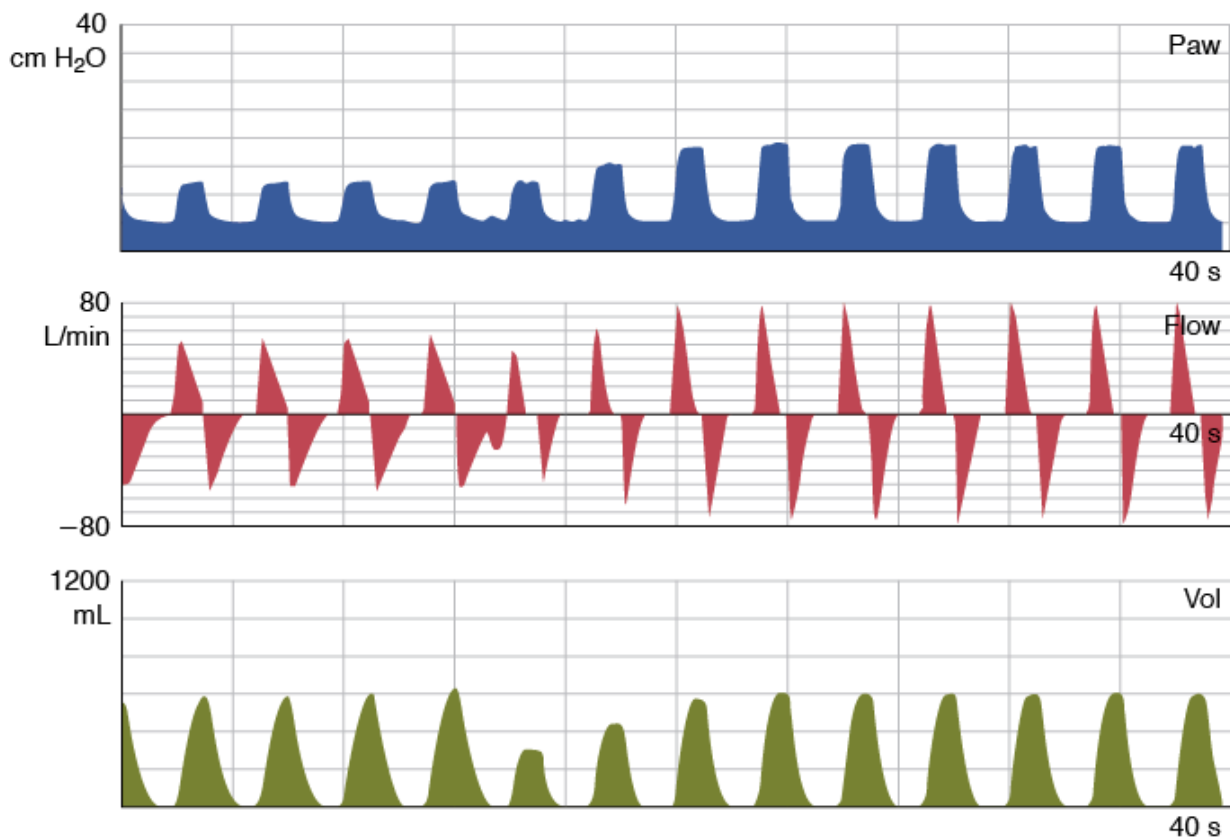
Dual Control Breath to Breath

DCBB techniques use standard pressure-targeted breaths (either pressure support or PACV), but the ventilator has the ability to adjust the pressure target according to a clinician-set V_T and the delivered V_T of previous breaths.^{13,14,23–26} When DCBB breaths are exclusively supplied with time cycling, the mode is commonly referred to as pressure-regulated volume control, although there are a number of proprietary names (e.g., Dräger’s AutoFlow, Covidien’s VC+, Hamilton’s Adaptive Pressure Ventilation). When DCBB breaths are supplied exclusively with flow cycling, the mode is commonly referred to as volume support. Some ventilators will switch between these two breath types depending on the number of patient efforts (e.g., Maquet’s Automode). With all of these modes, instead of setting a target pressure, the clinician selects

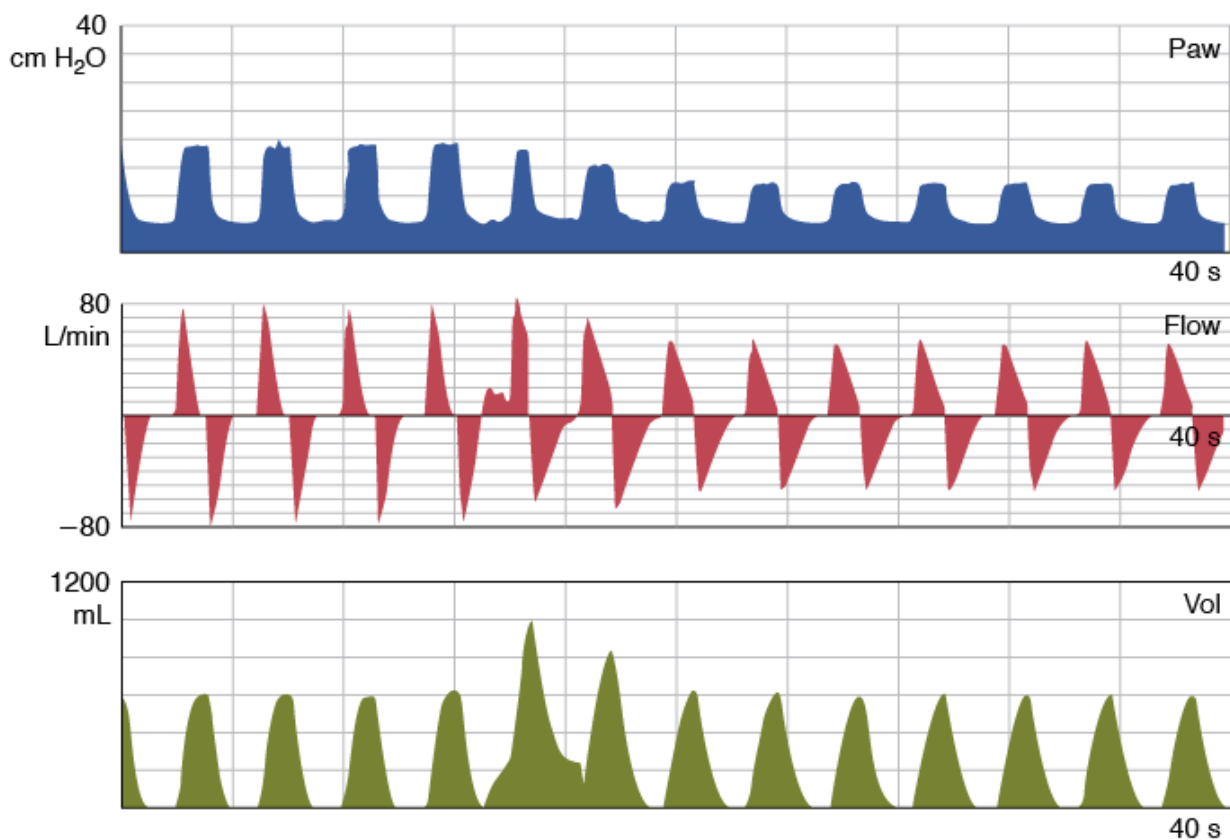
a V_T . The ventilator then delivers one or more “test breaths” with a small amount of inspiratory pressure. The V_T exiting the ventilator is measured, and total respiratory system compliance is calculated. Thereafter, each subsequent breath uses the previous calculation of system compliance to manipulate the ensuing pressure target to achieve the desired V_T (Fig. 15-1). The maximum pressure change from breath to breath on most systems generally is limited to a few centimeters of water (<3 cm H_2O) to prevent large swings in pressure and volume. The volume signal used for DCBB feedback control is not exhaled V_T but V_T exiting the ventilator. This prevents a runaway effect, which could occur if a leak in the circuit prevented accurate measurement of exhaled V_T .

Figure 15-1

Upper panel



Lower panel



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Behavior of DCBB breaths in a lung model simulation of changing lung compliance. In both the *top* and *bottom panels*, pressure (*Paw*), flow, and volume (*Vol*) are plotted over time. The target tidal volume in both panels is 600 mL. In the *top*

panel, lung compliance decreases after the fourth breath. Initially, there is a drop in tidal volume, but then the DCBB algorithm gradually increases the target inspiratory pressure to restore the volume. In the *bottom panel*, lung compliance increases after the fourth breath. Initially, there is an increase in tidal volume, but then the DCBB algorithm gradually reduces the inspiratory pressure target to restore the volume. (Reproduced, with permission, from Branson et al: The role of ventilator graphics when setting dual-control modes, *Respir Care* Feb;50(2); 187-201; 2005.)

In volume support, the flow-cycling criterion is either manufacturer-specific (e.g., 25% to 35% of peak flow) or, on many newer machines, clinician-adjustable. A secondary cycling mechanism may be present on some devices if inspiratory time exceeds a certain fraction (e.g., 80%) of a set total cycle time. Also, as with other pressure-targeted breaths, a rate-of-rise adjustment usually is available on DCBB breaths. When time-cycled DCBB breaths are interspersed with either spontaneous unsupported or pressure-supported breaths, or when flow-cycled DCBB breaths are interspersed with conventional PACV breaths, algorithms similar to SIMV or mandatory minute volume are often used to determine which breath type will be delivered.²⁷

Both animal and human studies show that DCBB breaths function as designed using both flow-cycled and time-cycled approaches.²³⁻²⁹ These studies emphasize that the DCBB breath is similar to other pressure-targeted breaths with enhanced gas mixing and better patient-ventilator flow synchrony as compared with flow-targeted, volume-cycled breaths. They also confirm that DCBB breaths do provide a volume guarantee without untoward side effects. Not surprisingly, several of these studies showed lower peak pressures with DCBB breaths than with volume assist-control breaths.^{23,24} This, however, is a finding consistent with the decelerating flow patterns of all pressure-targeted breaths (e.g., pressure support and PACV).

The role of DCBB modes in providing “lung-protective strategies” for patients with acute lung injury is unclear. Lung-protective strategies are based on data from numerous animal studies and clinical trials that low tidal volumes and low end-inspiratory lung-stretching pressures reduce ventilator-induced lung injury.³⁰ Setting a low tidal volume (e.g., 6 mL/kg ideal body weight) directly with flow-targeted volume-cycled breaths is the most reliable way to assure that a desired low tidal volume is delivered. Some researchers argue, however, that the set flow breath during low tidal ventilation may be particularly uncomfortable and that the variable flow feature of pressure-targeted breaths may synchronize better with patient efforts under these circumstances.^{31,32} Assuring a guaranteed tidal volume with a pressure-targeted DCBB breath would seem an attractive way to combine these features. Kallet et al addressed this issue. Although they found that DCBB breaths did provide a more reliable small tidal volume ventilatory pattern than pure pressure assist control, in a minority of patients up to 14% of tidal volumes were above the desired target value.^{33,34} Whether this variability is an acceptable trade-off for improved comfort during lung-protective ventilation needs further study.

DCBB breaths have also been evaluated during the ventilator-withdrawal process. Several evidence-based consensus groups have found that routine use of daily spontaneous breathing trials is critical in assessing ventilator discontinuation potential.³⁵ Less clear, however, is the role of gradually reducing ventilator support (“weaning”), either before or between spontaneous breathing trials. If weaning is desired, however, the DCBB modes have some conceptual appeal. Theoretically, the DCBB breath could be used to automatically reduce applied inspiratory pressure as the patient’s ability to breathe improved. Conversely, inspiratory pressure would increase if patient effort diminished or respiratory system mechanics worsened. Flow-cycled DCBB breaths, either alone or in combination with time-cycled DCBB breaths or various SIMV and/or pressure-support modes, have been used in several weaning studies. In general, they have performed as well as (or sometimes better than) standalone SIMV or pressure-support protocols, especially in the rapidly

recovering (e.g., postoperative) patient.^{29,36–39} A common finding in these weaning studies is that the DCBB breath modes required fewer ventilator manipulations. One must be cautious, however, in interpreting these weaning studies. A number of clinical trials show that the SIMV or SIMV + pressure-support control strategies delay weaning inappropriately as compared with spontaneous breathing trials or stand-alone pressure-support strategies.^{40,41} A more appropriate evaluation of DCBB breath-weaning strategies would be a comparison with spontaneous breathing trials delivered according to a protocol.^{35,42,43}

Unfortunately, the simplicity of the DCBB modes for weaning may produce problems.⁴⁴ For instance, if the clinician-set volume is excessive for patient demand, a recovering patient may not attempt to take over the work of breathing for that volume and thus support reduction and weaning may not progress. In addition, if the pressure level increases in an attempt to maintain an inappropriately high set tidal volume in the patient with airflow obstruction, intrinsic PEEP may result. On the other hand, a patient may receive inadequate support if the clinician-set tidal volume is inadequate for patient demand. Under these conditions, a patient will perform excessive work to maintain a patient-desired tidal volume while the inspiratory pressure is being reduced because volume exceeds the clinician setting. Clinicians need to be aware of the behavior of DCBB breaths under a variety of circumstances to properly use this mode.

Tidal Volume Feedback Modes that Enhance the Dual Control Breath-to-Breath Principle

Airway occlusion pressure ($P_{0.1}$),⁴⁵ oxygen saturation (SpO_2),^{37–39} and end-tidal CO_2 concentrations^{46,47} have been incorporated into DCBB mode-control algorithms to adjust either the target V_T or the breath-delivery pattern. The one system that is commercially available uses end-tidal CO_2 and respiratory rate along with the tidal volume to adjust the applied inspiratory pressure.⁴⁷ Known by the proprietary trade name *SmartCare* (Dräger), the computerized feedback system attempts to find an inspiratory pressure support that maintains the respiratory rate and tidal volume in a clinician-set “comfort zone.” The end-tidal CO_2 serves as a backup signal to assure adequate ventilation. The system is designed to wean the inspiratory pressure to as low a level as possible within these boundaries and then alert the clinician to perform a spontaneous breathing trial when this pressure reaches 9 cm H_2O .

A number of small observational trials have been done showing that the system did, indeed, keep patients in the clinician-selected “comfort zone” for 95% of the time.^{46,47} In a larger randomized clinical trial, this approach appeared to remove ventilator support quicker than “physician-controlled” weaning.⁴⁸ Unfortunately, this control group did not have a protocolized spontaneous breathing trial approach, and thus may have had support removal delayed. Indeed, a subsequent trial noted that in the presence of an active protocol for discontinuing mechanical ventilation, automated withdrawal offered no advantage.⁴⁹ Even if it is not superior, however, an automated system that is “just as good” as clinicians could have applications in settings with rapidly recovering patients or low availability of clinicians to make frequent assessments.

Feedback Control of Ventilator Breath Delivery Based on Respiratory System Mechanics

A novel approach to automated feedback control of ventilator support combines the DCBB principle with an integrated V_T , frequency and inspiratory-to-expiratory (I:E) ratio algorithm based on respiratory system mechanics. Known as *adaptive lung ventilation* or *adaptive-support ventilation* (ASV, Hamilton Medical),^{50–55} the breath-control algorithm

attempts to partition the frequency, tidal volume, and I:E ratio so as to minimize ventilator–patient inspiratory work and intrinsic PEEP. ASV does this by calculating respiratory system mechanics using several “test breaths.” It then uses a “minimal work” calculation to set the frequency–tidal volume pattern that minimizes the combined resistance and compliance components of work. The clinician must input the patient’s predicted body weight and the percent of desired minute ventilation. The predicted body weight frames the tidal volume range, avoiding volutrauma and hypoventilation. The predicted needed minute volume is 0.1 L/kg/min. As an example, a 70-kg patient set at 100% would receive a 7 L minute volume. If the percent minute volume was set to 125%, in the presence of increased physiologic dead space the minute ventilation would be 8.75 L. The ASV algorithm then uses a measurement of the expiratory time constants (RC_e = resistance $\times$ compliance) to ensure an inspiratory time of at least one RC_e and an expiratory time of at least three RC_e s. The actual formula is:

$$f = \frac{\sqrt{1 + 2aRC(\dot{V}_A/V_D)} - 1}{aRC} \quad (1)$$

where RC is the respiratory time constant (the product of resistance R and compliance C), $\dot{V}_A$ and V_D are alveolar ventilation and dead space ventilation, respectively, and a is a constant that depends on the flow waveform.

Boundary rules exist to prevent excessive (runaway) settings ([Table 15-1](#)). Clinicians must set the desired minute ventilation and the proportion of that minute ventilation that the machine is to supply. Ideal body weight also can be used to calculate the desired minute ventilation based on metabolic demands and predicted dead space. Clinicians also must set the PEEP and FI_{O_2} .

Table 15-1: Boundary Conditions for Adaptive Lung or Support Ventilation

Parameter	Minimum	Maximum
Inspiratory pressure (cm H ₂ O)	5 above baseline airway pressure (PEEP/CPAP)	10 below P _{max} alarm setting
Tidal volume (mL)	4.4 · IBW	15.4 · IBW or $\dot{V}_E/5$, whichever is lower (may be limited by P _{max} alarm)
Target respiratory rate (bpm)	5 bpm	22 bpm · % min vol/100 (if IBW > 15 kg)
		45 bpm · % min vol/100 (if IBW < 15 kg)
Mandatory breath rate (bpm)	5 bpm	60 bpm
Inspiratory time (s)	0.5 s or 1 · RCe, whichever is longer	2 s
Expiratory time (s)	2 · RCe (possibly 3 · RCe)	12 s
I:E ratio range	1:4	1:1

% min vol, clinician-set proportion of predicted minute volume needed by the patient that will be supplied by the ventilator (the percent minute volume is based on predicted body weight, e.g., for an 80-kg patient, 100% of minute volume is a minute ventilation of 8 L/min); bpm, allowable breaths/min; CPAP, continuous positive airway pressure; IBW, ideal body weight; I:E, inspiratory-to-expiratory-time ratio; PEEP, positive end-expiratory pressure; P_{max}, clinician-set maximal inspiratory pressure; RCe, resistance × compliance; $\dot{V}_E$, exhaled minute volume.

Note: These parameters set the limits on the various parameters used during ASV.

ASV as a pure control mode has been evaluated in a number of ways. Initial lung-model testing⁵⁶ demonstrated that the ASV algorithm responded properly to abrupt changes in lung mechanics. Several early clinical studies have compared initial ASV settings with traditional clinician-selected settings and have found that ASV tends to select a lower tidal volume and faster rate (and thus lower inspiratory pressures) than do clinicians.^{56–59} Two other early studies suggest that ASV also appropriately adapts to changes in patient position and the change from double- to single-lung ventilation during anesthesia.^{55–61} One other study suggested that the I:E algorithm of ASV produced less air trapping in patients with chronic obstructive pulmonary disease.⁶² Longer-duration clinical studies with ASV show that the algorithm provides adequate ventilator support in anesthetized patients,^{56,59} as well as in patients with respiratory failure.⁶³

More recent evaluations of ASV, however, have focused on its ability to provide appropriate lung-protective small tidal volumes. Indeed, when respiratory system compliance is poor, the ASV algorithm supplies a protective low tidal volume ventilator pattern similar to that recommended by the ARDS Network.⁵⁴ Problems arise, however, when respiratory system compliance is less deranged (e.g., patients with milder forms of acute lung injury). Under these conditions, the

ASV algorithm tends to deliver tidal volumes often in excess of 10 mL/kg ideal body weight.^{45,64} The clinical significance of this is unknown but the potential harm from this should be considered by clinicians wishing to use this mode.

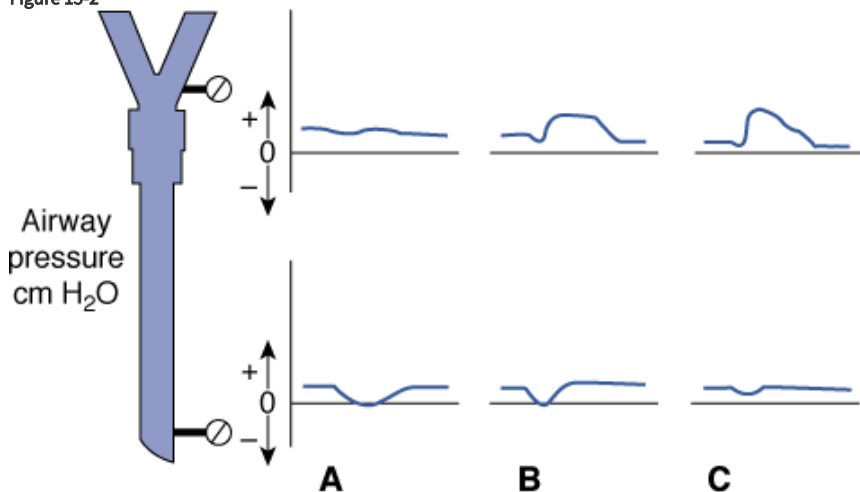
ASV also might be considered an automatic weaning mode because the algorithm responds with lower pressures and fewer mandatory breaths as patient effort increases. When spontaneous efforts occur with ASV, the algorithm continues to try to conform to the minimal work tidal volume considerations above and in that sense resembles the feedback features of volume support noted above.⁶⁵ The ASV feedback control, however, is more complex than volume support in that the predicted body weight input as well as respiratory system resistance, and compliance (and the resulting time constant) modulate the tidal volume target. A number of studies have evaluated ASV in patients being weaned from mechanical ventilation.^{50,63,66–70} In general, these studies show that ASV safely provides adequate ventilator support and has similar (or faster) weaning times as compared with various SIMV and SIMV + pressure support protocols.^{66,70} These studies also generally show fewer ventilator manipulations with ASV. As noted earlier for other DCBB approaches, a more appropriate evaluation of ASV weaning strategies would be a comparison with use of spontaneous breathing trials delivered by protocol,^{35,42,43} not with SIMV or SIMV + pressure support.

Larger trials in patients with different forms of lung injury clearly are needed to establish the appropriateness of the ASV algorithms in various disease states and the effects of ASV on outcome.

Automatic Adjustments in Pressure and Flow Based on Artificial Airway Geometry

The endotracheal tube (ETT) imposes a significant inspiratory resistance on a spontaneously breathing patient.^{71,72} This imposed load can have an impact on flow synchrony during interactive assisted or supported breaths, and can make it difficult to assess potential for ventilator withdrawal during periods of unassisted or unsupported breathing (Fig. 15-2, *left panel*).

Figure 15-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Effects of the endotracheal tube (ETT) resistance on the application of continuous positive airway pressure (CPAP), pressure support, and CPAP with ETT automatic tube compensation (CPAP-ATC) in a spontaneously breathing patient. Depicted are airway pressure profiles over time in the ventilator circuit (*upper panels*) and the distal trachea (*lower panels*). In the left breath, the CPAP tracing in the ventilator circuit is appropriately constant, whereas the tracheal

pressure downswing reflects the patient work required to effect flow through the ETT. In the middle breath, pressure support is applied in the ventilator circuit. This reduces some of the patient work, but because the “square wave” pressure-support breath does not adequately pressurize the trachea rapidly enough, there still can be significant work required by the patient early in the breath. In the right breath, a similar pressure-support breath is applied, but the pressure profile is altered to provide a higher pressure early and the target pressure later (CPAP-ATC). This profile is machine determined, and knowledge of ETT geometry accounts more appropriately for the effects of ETT resistance. It thus creates a truer CPAP pattern in the trachea. Note that this ETT compensation feature also can be combined with pressure support on some ventilators. (Reproduced, with permission, from MacIntyre: Respiratory mechanics in the patient who is weaning from the ventilator, *Respir Care* Feb;50(2); 275-86; 2005.)

Low-level (e.g., 5 to 8 cm H₂O) pressure support has been proposed as a way of eliminating the ETT resistive load.^{73–75} Unfortunately, the pressure-support algorithm supplies a constant inspiratory pressure, which, because of the high fixed resistance of the ETT, tends to undercompensate the load at the beginning of the breath (see Fig. 15-2, middle panel). Patient muscle unloading thus is uneven and may be suboptimal.

One way to address this is to use a pressure sensor at the distal end of the ETT to target the pressure support applied pressure. This approach would provide a more even pressure application to the contracting inspiratory muscles. Unfortunately, this approach is unreliable because intra-air-way sensors are subject to errors from positioning and mucus occlusion. An alternative is to have the ventilator calculate the ETT resistance properties and use those calculations to manipulate applied pressure in such a way as to compensate for the ETT effects (see Fig. 15-2, right panel).^{76,77} To accomplish this, the clinician must input the tube geometry (length and diameter) and the percentage of compensation desired (10% to 100%). The ventilator then uses these data to calculate ETT resistance and incorporates this with measurements of instantaneous flow to apply pressure proportional to resistance throughout the total respiratory cycle.

The ETT compensation concept was first introduced in 1993 by Guttman et al⁷⁶ and was applied during both inspiration and expiration. It was believed that the expiratory effect (i.e., circuit pressure actually going below set baseline pressure or “subatmospheric” to “assist” expiratory flow) might further improve patient-ventilator interactions by reducing expiratory work and hyperinflation. In a series of ventilated patients, this approach was shown to improve subjective patient comfort substantially.⁷⁸

Variations on this computational approach to manipulating applied pressure are now available as automatic tube compensation (Dräger, Covidien, and Maquet) and automatic airway compensation (CareFusion). Bench and clinical studies verify that the ETT compensation algorithms on these commercial systems generally perform as designed during both continuous positive airway pressure and pressure support.^{79–81} Not all systems, however, have the expiratory feature, and the response characteristics of commercial systems may not be as robust as the Guttman design.⁷⁶

It also must be recognized that the ETT compensation strategy is based on the input geometry of the artificial airway and cannot account for changes in tube characteristics induced by kinks or partial occlusions or the relationship of the tube opening against the tracheal wall. Thus, this strategy should not be considered a perfect surrogate for a tracheal pressure sensor, especially as the duration of intubation increases. Nevertheless, despite these limitations, and despite the fact that ETT compensation has not been subjected to outcome studies, the simplicity of the concept and safety of the design would seem to warrant its application during pressure support and continuous positive airway pressure under most circumstances.

An interesting extension of the ETT compensation concept is to use a pleural pressure to target the positive-pressure breath.⁸² This approach requires an esophageal balloon to measure esophageal pressure, a surrogate for pleural pressure. Conceptually, a pleural pressure target could be used to control the breath directly in such a way as to account for all the components of patient work, not just that caused by the ETT. At the present time, this approach has been explored only with prototype systems in experimental animals.

Feedback Systems Controlling Positive End-Expiratory Pressure and Concentration of Fractional Inspired Oxygen

Maintenance of normal arterial [oxygen](#) saturation (SaO_2) is a key goal of mechanical ventilation. Hypoxemia has well-known consequences, including tissue injury, and if left uncorrected, can result in death. In modern intensive care units, hypoxemia is prevented through a variety of mechanisms including appropriate ventilator settings, sedation, analgesia, and maintenance of homeostasis. The current standard of care for detecting and ameliorating hypoxemia, however, has remained unchanged over the last 30 years. The clinician sets the low arterial oxyhemoglobin saturation (SpO_2) alarm setting and if the SpO_2 falls below this value (commonly 88%), an audible and visual alarm sounds. The alarm must be recognized by the clinician who must enter the room, elucidate the cause (e.g. ventilator disconnect, agitation, worsening pulmonary status), remedy the problem, and possibly increase the FI_{O_2} . This is an antiquated method that is analogous to trying to maintain the heat in your home by turning the furnace on or off based on falling below the “too cold” threshold.

Concentration of Fractional Inspired Oxygen Feedback Systems

The incidence of hypoxemia in ventilated adult intensive care unit patients is unknown. In neonatal intensive care units, hypoxemic events are frequent and life-threatening. Hyperoxemia also carries significant risk in this population. Recent data demonstrate that abdominal muscle contraction during crying and agitation is the most common cause of neonatal hypoxemia. The use of pressure-limited ventilation and uncuffed endotracheal tubes in infants, coupled with diaphragmatic contraction, profoundly limit [oxygen](#) delivery.⁸³ Both lung injury from [oxygen](#) toxicity and the retinopathy of prematurity are known complications of excessive [oxygen](#) exposure.

Closed-loop control of FI_{O_2} using either indwelling partial pressure of arterial [oxygen](#) (PaO_2) sensors, transcutaneous [oxygen](#) monitoring, and pulse oximetry has been attempted numerous times in the last 35 years.^{84–91} Most of these investigations met with limited success owing to the sensor response and time required to alter delivered FI_{O_2} . Several of these studies used simple [oxygen](#) delivery to a head hood where both increases and decreases in FI_{O_2} occur slowly.

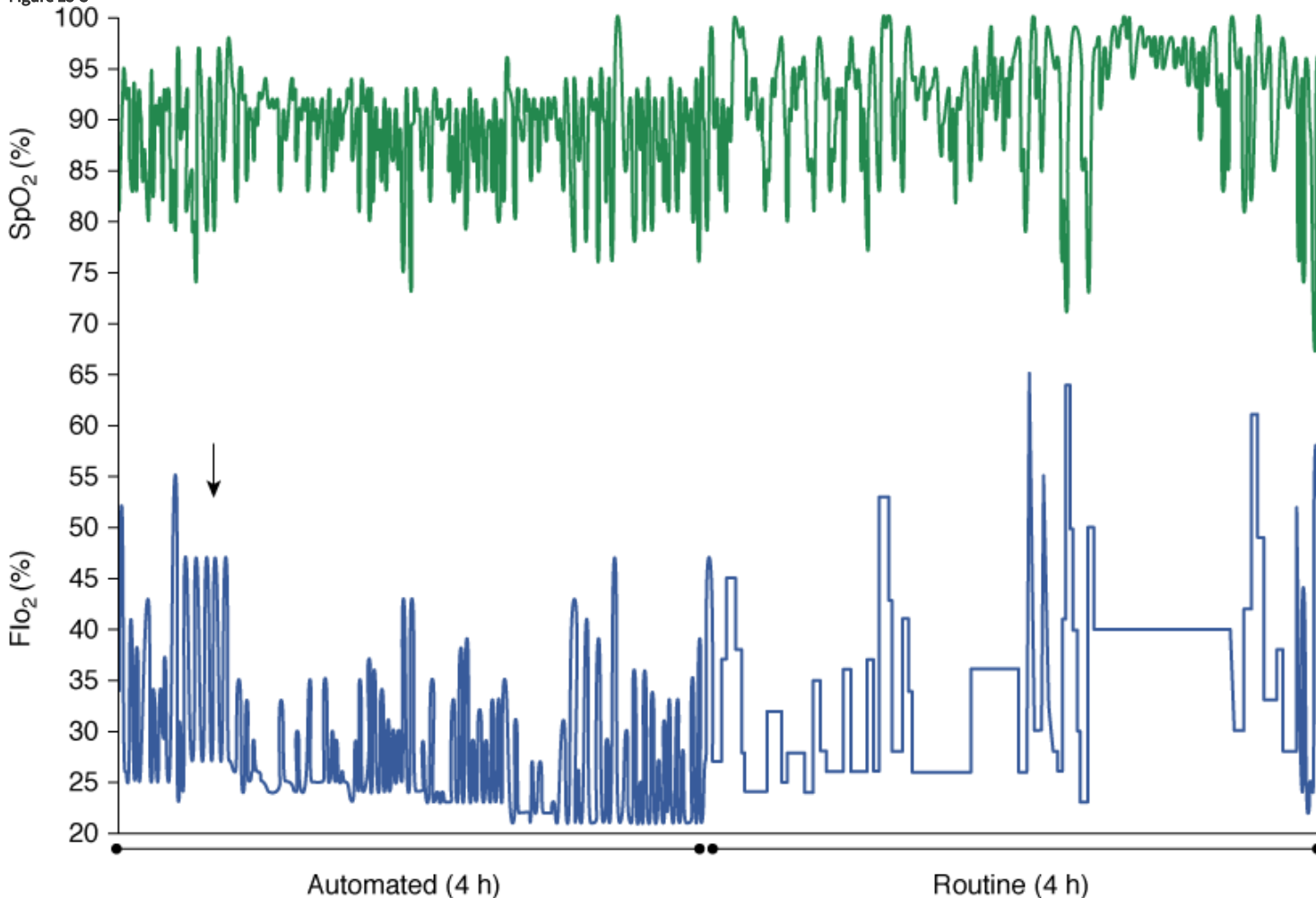
More recently, Claire et al performed a series of important investigations in closed-loop control of FI_{O_2} in neonates.^{92–95} In their initial studies, the frequency of hypoxemia was not different between manual FI_{O_2} adjustment and automated FI_{O_2} adjustment. The nursing workload, however, was significantly less with the automated system.⁹⁶ In a second trial of sixteen mechanically ventilated infants, they found that, compared with manual adjustment, the percentage of time in the target SpO_2 range (88% to 95%) was greater during the automated control of ($58\% \pm 10\%$ vs. $42\% \pm 9\%$).⁹³ The percentage of time with SpO_2 less than 88%, however, increased during the automated period ($33\% \pm 7\%$ vs. $27\% \pm 9\%$)

because of more frequent episodes of hypoxemia. The 4-hour median FI_{O_2} was lower during the automated period ($29\% \pm 4\%$ vs. $34\% \pm 5\%$).

The data suggest that by minimizing hyperoxemia (associated with retinal injury in premature infants), the incidence of hypoxemia may increase. The rapid response of the automated system to the hypoxemia, however, resulted in a similar durations of low SpO_2 between groups; that is, the number of hypoxemic events were more frequent, but the duration of each was shorter.

In their most recent trial of thirty-two premature infants, Claure et al demonstrated similar findings. The proportion of time that SpO_2 was in the target range (SpO_2 of 87% to 93%) was significantly greater during automated control of FI_{O_2} versus manual control ($40\% \pm 14\%$ vs. $32\% \pm 13\%$). Time with SpO_2 of less than 87%, however, increased significantly during the automated period ($32\% \pm 12\%$ vs. $23\% \pm 9\%$), with more frequent episodes with SpO_2 between 80% and 86%, whereas times with SpO_2 of less than 80% or less than 75% were not different. They also demonstrated significantly fewer manual FI_{O_2} changes (10 ± 9 vs. 112 ± 59 changes per 24 hours), compared with the manual period.⁹⁵ Figure 15-3 demonstrates the frequency of low SpO_2 events and required FI_{O_2} corrections in a single neonate on mechanical ventilation during a 4-hour period. This system remains under investigation and is not commercially available.

Figure 15-3



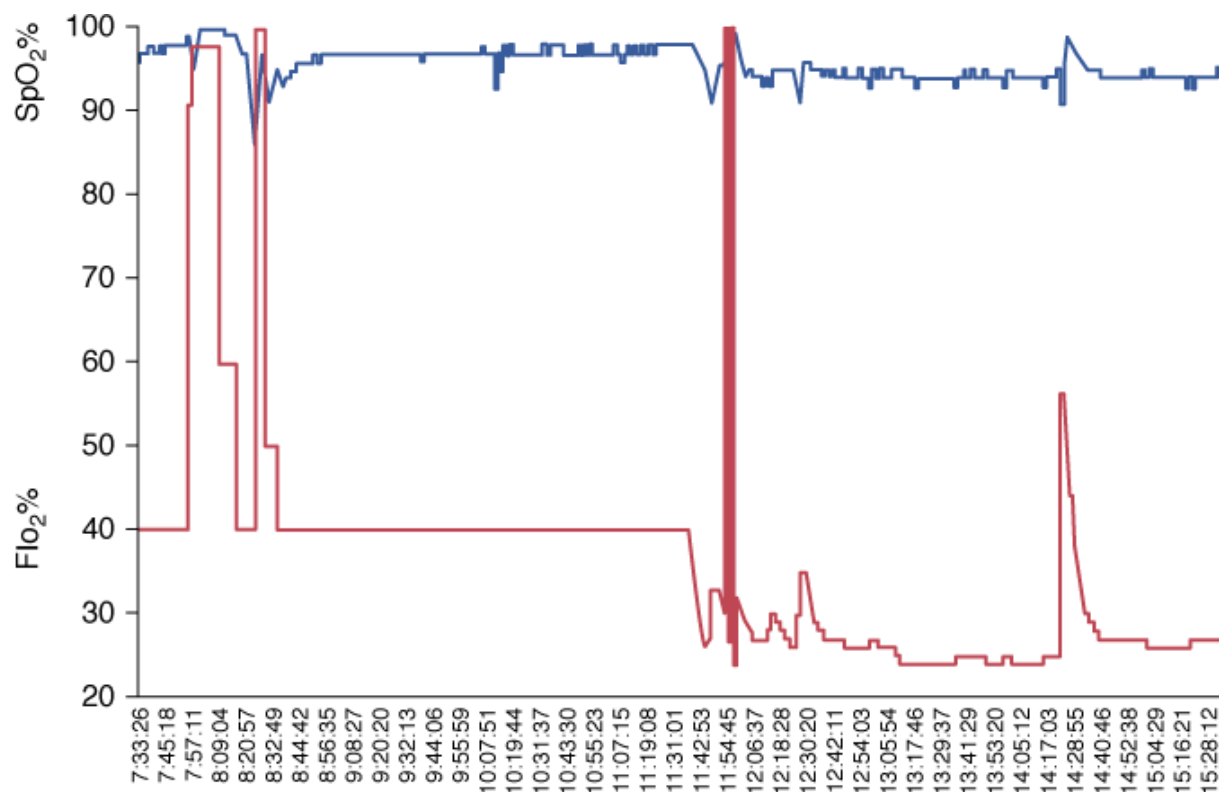
This recording of SpO₂ and FI_{O2} illustrates the difficulty in maintaining SpO₂ within the intended range by manual FI_{O2} adjustments during the routine period. These adjustments were not always consistent, and exposure to high FI_{O2} was at times prolonged. Automated FI_{O2} increases were brief in a gradually decreasing basal FI_{O2}. The *arrow* points to manual FI_{O2} adjustments during care procedures with the automated system on standby. (Used, with permission, from Claure, D'Ugard, and Bancalari.⁹³)

Johannigman et al investigated closed-loop FI_{O2} in adult trauma patients with the goals of achieving normoxemia, preventing hypoxemia, and conserving oxygen.^{97,98} These goals have significant importance during transport and in military operations. Closed-loop control of FI_{O2} provides normoxemia regardless of the skill of the caregiver and prevents hypoxemia when patients are unattended.

Oxygen conservation is rarely an issue in the hospital, but is critical in resource-limited environments (e.g., mass casualty, far-forward deployments).

In an early pilot and follow-up study of forty-five trauma patients, Johannigman et al compared 4 hours of closed-loop control to 4 hours of manual FI_{O2} control. They found that the target SpO₂ (92% to 96%) was achieved 82% of the time during closed-loop control versus only 37% of the time during manual control. The most common “out-of-target” condition in the manual control group was hyperoxemia (SpO₂ >97%), seen 61% of the time versus 17% during closed-loop control. The total time spent with SpO₂ less than 88% per patient was small (<1 minute) and was not different between groups. The authors also noted that many trauma patients never manifest hypoxemia, regardless of the study arm. Figure 15-4 is a graph of FI_{O2} and SpO₂ during the 8-hour trial. This finding was recently supported by a Dutch study demonstrating hyperoxemia in trauma patients in spite of low inspired oxygen (<0.40).⁹⁹ These authors demonstrated that even in the face of significant hyperoxemia (PaO₂ >120 mm Hg), if inspired oxygen was equal to or less than 0.40, clinicians rarely reduced FI_{O2}. This was thought to be secondary to the commonly held belief that FI_{O2} less than 0.50 is nontoxic to the lungs. This, however, does not account for potential negative consequences at the cellular level as a result of oxidative stress.¹⁰⁰ Oxygen usage was significantly less during closed-loop control (3 L/min) versus manual control (2.1 L/min). This 33% decrease in oxygen usage might be significant where oxygen stores are limited. Additionally, this level of oxygen requirement could be met with an oxygen concentrator.

Figure 15-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Example of closed-loop control of SpO₂ in an adult patient with traumatic respiratory failure. The first 4 hours represent the manual control period and the second 4 hours the automatic control period. Increases in FiO₂ to 100% are associated with hypoxemia (SpO₂ <88%).

Both these systems use a proportional-integral-derivative (PID) controller. In this case, a negative feedback PID system. A negative feedback system attempts to minimize the difference between the target value and the actual value. Ideally, the difference between target and measured is zero. In this instance, the FiO₂ is changed until the difference between the target SpO₂ and the measured SpO₂ is zero. A PID controller allows the FiO₂ to be changed in large steps based on the level of SpO₂ and the difference in set and actual SpO₂. When the target and measured values are close, the changes in FiO₂ are smaller. The FiO₂ can be adjusted both by changing the stepwise increase or decrease as well as the time between steps. At the time of this writing, no closed-loop FiO₂ system for mechanical ventilation is commercially available.

Concentration of Fractional Inspired Oxygen and Positive End-Expiratory Pressure Feedback Systems

Setting PEEP remains one of the most contentious issues in mechanical ventilation. Although automated control of FiO₂ is relatively simple, the effects of PEEP on hemodynamics, airway pressures, and lung protection create a much greater challenge. In preliminary findings, Lellouche et al found that in cardiac surgery patients, automated control of PEEP and FiO₂ resulted in similar outcomes compared to manual manipulations.¹⁰¹ Interestingly, this group utilized the PEEP-FiO₂ table from the original ARDS Network trial to control the algorithm. We believe that a commercially available version of closed-loop control of PEEP would require some inference of hemodynamic performance or confirmation of hemodynamic stability by the caregiver. It may be that a decision-assist system for PEEP would be the first step.

Feedback Systems Driven by Novel Sensors of Patient Effort

In the last decade, two novel feedback modes for delivering positive pressure in response to patient effort have been introduced.⁵⁰ The first is proportional-assist ventilation (PAV), which uses measured patient-generated inspiratory flow and then puts a clinician-set gain on both pressure and flow delivery.¹⁰² PAV needs estimates of respiratory system compliance and resistance, which are determined from intermittent controlled “test” breaths or periodic pauses randomly applied during ventilation. The only clinician setting is then a proportion of the calculated work of breathing that the delivered flow and pressure are meant to overcome. The second is neurally adjusted ventilator assist (NAVA), which uses a diaphragmatic electromyogram signal on an esophageal catheter to determine patient effort during all three phases of breath delivery (i.e., trigger, flow delivery, and cycling).¹⁰³ Like PAV, NAVA applies a clinician-set gain to provide pressure and flow adjustments in proportion to the strength of the electromyogram signal. As these modes are not considered “conventional” modes of ventilation, they are not discussed further here. The reader is referred to more detailed accounts of these approaches in [Chapters 12](#) and [13](#).

Conclusion

A positive-pressure breath ideally should provide a V_T that is adequate for gas exchange and appropriate muscle unloading while minimizing any risk for injury or discomfort. The latest generation of ventilators is now using sophisticated feedback systems to “sculpt” positive-pressure breaths according to patient effort and respiratory system mechanics. At the present time, however, these new control strategies are not totally “closed-loop” systems because the automatic input variables are still limited, some clinician settings are still required, and the specific features of the “perfect” breath design still are not entirely clear. Despite these limitations, there is at least some rationale for many of these newer feedback features, even though all of them await outcome studies to justify their widespread use.

References

1. MacIntyre NR. Evolution of the modern mechanical ventilator. In: Hill N, Levy M, eds. *Ventilator Management Strategies for Critical Care*. New York, NY: Marcel Dekker; 2000;1–20.

2. Luce JM, Pierson DJ, Hudson LD. Intermittent mandatory ventilation. *Chest*. 1981;79:678–685. [[PubMed: 6785016](#)]

3. Weisman IM, Rinaldo JE, Rogers RM, et al. Intermittent mandatory ventilation. *Am Rev Respir Dis*. 1983;127:641–647. [[PubMed: 6342483](#)]

4. Heenan TJ, Downs JB, Douglas ME, et al. Intermittent mandatory ventilation: is synchronization important? *Chest*. 1980;77:598–602. [[PubMed: 6767582](#)]

5. Hewlett AM, Platt AS, Terry VG. Mandatory minute volume: a new concept in weaning from mechanical ventilation. *Anaesthesia*. 1977;32:163–169. [[PubMed: 322535](#)]

6. Branson RD, Chatburn RL. Technical description and classification of modes of ventilator operation. *Respir Care*. 1992;37:1026–1044. [[PubMed: 10183731](#)]

7. MacIntyre NR. Principles of mechanical ventilation. In: Murray J, Nadel J, eds. *Textbook of Respiratory Medicine*, 5th ed. New York, NY: Elsevier; 2010;2084–2103.

8. MacIntyre NR, Branson RD, eds. *Mechanical Ventilation*. 2nd ed. Philadelphia, PA: Elsevier; 2009.

9. MacIntyre NR. Respiratory function during pressure support ventilation. *Chest*. 1986;89:677–683. [[PubMed: 3698697](#)]

10. MacIntyre NR, McConnell R, Cheng KC, et al. Patient-ventilator flow dyssynchrony: flow-limited versus pressure-limited breaths. *Crit Care Med*. 1997;25:1671–1677. [[PubMed: 9377881](#)]

11. Kallet RH, Campbell AR, Alonso JA, et al. The effects of pressure control versus volume control assisted ventilation on patient work of breathing in acute lung injury and acute respiratory distress syndrome. *Respir Care*. 2000;45:1085–1096. [[PubMed: 10980100](#)]

12. Branson RD, Johannigman JA. Role of ventilator graphics when setting dual control modes. *Respir Care*. 2005;50:187–201. [[PubMed: 15691391](#)]

13. Branson RD, MacIntyre NR. Dual control modes of mechanical ventilation. *Respir Care*. 1996;41:294–305.

14. Branson RD, Davis K. Dual control modes: combining volume and pressure breaths. *Respir Care Clin N Am*. 2001;7:397–401. [[PubMed: 11517030](#)]

15. Raneri VM. Optimization of patient ventilator interactions: closed loop technology. *Intensive Care Med*. 1997;23:936–939.

16. Amato MBP, Barbos CSV, Bonassa J, et al. Volume assisted pressure support ventilation (VAPSV): a new approach for reducing muscle workload during acute respiratory failure. *Chest*. 1992;102:1225–1234. [[PubMed: 1395773](#)]

17. Marini JJ, Rodriguez RM, Lamb V. The inspiratory workload of patient-initiated mechanical ventilation. *Am Rev Respir Dis*. 1986;134:902–909. [[PubMed: 3096177](#)]

18. Marini JJ, Capps JS, Culver BH. The inspiratory work of breathing during assisted mechanical ventilation. *Chest*. 1985;87:612–618. [[PubMed: 3987373](#)]

19. Knebel AR, Janson-Bjerklie SL, Malley JD, et al. Comparison of breathing comfort during weaning with two ventilatory modes. *Am J Respir Crit Care Med*. 1994;149:14–18. [[PubMed: 8111572](#)]

20. Cinnella G, Conti G, Lofaso F, et al. Effects of assisted ventilation on the work of breathing: volume-controlled versus pressure-controlled ventilation. *Am J Respir Crit Care Med*. 1996;153:1025–1033. [[PubMed: 8630541](#)]

21. Kallet RH, Campbell AR, Alonso JA, et al. The effects of pressure control versus volume control assisted ventilation on patient work of breathing in acute lung injury and acute respiratory distress syndrome. *Respir Care*. 2000;45:1085–1096. [[PubMed: 10980100](#)]

22. MacIntyre NR, Gropper C, Westfall T. Combining pressure limiting and volume cycling features in a patient-interactive mechanical ventilation. *Crit Care Med*. 1994;22:353–357. [[PubMed: 8306697](#)]

23. Piotrowski A, Sobala W, Kawczynski P. Patient initiated, pressure regulated, volume controlled ventilation compared with intermittent mandatory ventilation in neonates: a prospective, randomized study. *Intensive Care Med.* 1997;23:975–981. [[PubMed: 9347370](#)]
-
24. Kocis KC, Dekeon MK, Rosen HK, et al. Pressure-regulated volume control vs volume control ventilation in infants after surgery for congenital heart disease. *Pediatr Cardiol.* 2001;22:233–237. [[PubMed: 11343150](#)]
-
25. Guldager H, Nelso SL, Carl P, et al. A comparison of volume controlled ventilation and pressure regulated volume control in acute respiratory failure. *Crit Care.* 1997;1:75–77. [[PubMed: 11056699](#)]
-
26. D'Angio CT, Chess PR, Kovacs SJ, et al. Pressure regulated volume control vs. synchronized intermittent mandatory ventilation for very low birth rate infants: a randomized controlled trial. *Arch Pediatr Adolesc Med.* 2005;159:868–875. [[PubMed: 16143747](#)]
-
27. Roth H, Luecke T, Lansche G, et al. Effects of patient-triggered automatic switching between mandatory and supported ventilation in the postoperative weaning period. *Intensive Care Med.* 2001;27:47–51. [[PubMed: 11280672](#)]
-
28. Sottiaux TM. Patient-ventilator interactions during volume-support ventilation: asynchrony and tidal volume instability—a report of three cases. *Respir Care.* 2001;46:232–233.
-
29. Abubakar KM, Keszler M. Patient-ventilator interactions in new modes of patient-triggered ventilation. *Pediatr Pulmonol.* 2001;32:71–75. [[PubMed: 11416879](#)]
-
30. Dreyfuss D, Savmon G. Ventilator induced lung injury: lessons from experimental studies. *Am J Resp Crit Care Med.* 1998;157:294–323. [[PubMed: 9445314](#)]
-
31. Yang LY, Huang YC, MacIntyre NR. Patient-ventilator synchrony during pressure-targeted versus flow-targeted small tidal volume assisted ventilation. *J Crit Care.* 2007;22(3):252–257. [[PubMed: 17869978](#)]
-
32. MacIntyre NR, Sessler CN. Are there benefits or harm from pressure targeting during lung-protective ventilation? *Respir Care.* 2010;55(2):175–180. [[PubMed: 20105343](#)]
-
33. Kallet RH, Campbell AR, Dicker RA, et al. Work of breathing during lung-protective ventilation in patients with acute lung injury and acute respiratory distress syndrome: a comparison between volume and pressure-regulated breathing modes. *Respir Care.* 2005; 50:1623–1631 [[PubMed: 16318643](#)]
-
34. Cinnella G, Conti G, Lofaso F, et al. Effects of assisted ventilation on the work of breathing: volume-controlled versus pressure-controlled ventilation. *Am J Respir Crit Care Med.* 1996;153:1025–1033. [[PubMed: 8630541](#)]
-
35. ACCP/SCCM/AARC Task Force. Evidence based guidelines for weaning and discontinuing mechanical ventilation. *Chest.* 2001;120:375S–395S.
-
36. Randolph AG, Wypij D, Venkataraman ST, et al. Effect of mechanical ventilator weaning protocols on respiratory outcomes in infants and children: a randomized, controlled trial. *JAMA.* 2002;288:2561–2568. [[PubMed: 12444863](#)]
-

37. Strickland JH, Hasson JH. A computer-controlled ventilator weaning system. *Chest*. 1991;100:1096–1099. [[PubMed: 1914564](#)]
-
38. Tong DA. Weaning patients from mechanical ventilation: a knowledge-based system approach. *Comput Methods Programs Biomed*. 1991;35:267–278. [[PubMed: 1752122](#)]
-
39. Strickland JH, Hasson JH. A computer-controlled ventilator weaning system: a clinical trial. *Chest*. 1993;103:1220–1226. [[PubMed: 8131469](#)]
-
40. Esteban A, Frutos F, Tobin MJ, et al. A comparison of four methods of weaning patients from mechanical ventilation. *N Engl J Med*. 1995;332:345–350. [[PubMed: 7823995](#)]
-
41. Brochard L, Rauss A, Benito S, et al. Comparison of three methods of gradual withdrawal from ventilatory support during weaning from mechanical ventilation. *Am J Respir Crit Care Med*. 1994;150:896–903. [[PubMed: 7921460](#)]
-
42. Ely EW, Baker AM, Dunagan DP, et al. Effect on the duration of mechanical ventilation of identifying patients capable of breathing spontaneously. *N Engl J Med*. 1996;335:1864–1869. [[PubMed: 8948561](#)]
-
43. Kollef MH, Shapiro SD, Silver P, et al. A randomized, controlled trial of protocol-directed versus physician-directed weaning from mechanical ventilation. *Crit Care Med*. 1997;25:567–574. [[PubMed: 9142019](#)]
-
44. Jaber S, Delay JM, Matecki S, et al. Volume-guaranteed pressure-support ventilation facing acute changes in ventilatory demand. *Intensive Care Med*. 2005;31:1181–1188. [[PubMed: 16032424](#)]
-
45. Iotti GA, Braschi A. Closed loop support of ventilator workload: the P0.1 controller. *Respir Care Clin N Am*. 2001;7:441–451. [[PubMed: 11517033](#)]
-
46. Dojat M, Harf A, Touchard D, et al. Evaluation of a knowledge-based system providing ventilatory management and decision for extubation. *Am J Respir Crit Care Med*. 1996;153:997–1004. [[PubMed: 8630586](#)]
-
47. Dojat M, Harf A, Touchard D, et al. Clinical evaluation of a computer-controlled pressure support mode. *Am J Respir Crit Care Med*. 2000;161:1161–1166. [[PubMed: 10764306](#)]
-
48. Lellouche F, Mancebo J, Jolliet P, et al. A multicenter randomized trial of computer-driven protocolized weaning from mechanical ventilation. *Am J Respir Crit Care Med*. 2006;174:894–900. [[PubMed: 16840741](#)]
-
49. Rose L, Presneill J, Johnston L, Cade JF. A randomised, controlled trial of conventional versus automated weaning from mechanical ventilation using SmartCare/PS. *Intensive Care Med*. 2008;34:1788–1795. [[PubMed: 18575843](#)]
-
50. Lellouche F, Brochard L. Advanced closed loops during mechanical ventilation. *Best Pract Res Clin Anaesthesiol*. 2009;23:81–93. [[PubMed: 19449618](#)]
-
51. Campbell RS, Branson RD, Johannigman JA. Adaptive support ventilation. *Respir Care Clin N Am*. 2001;7:425–440. [[PubMed: 11517032](#)]
-

52. Brunner JX, Laubscher TP, Banner MJ, et al. A simple method to measure total expiratory time constant based on the passive expiratory flow volume curve. *Crit Care Med*. 1995;23:1117–1122. [[PubMed: 7774225](#)]
-
53. Iotti GA, Polito A, Belliato M, et al. Adaptive support versus conventional ventilation for total ventilator support in acute respiratory failure. *Intensive Care Med*. 2010;36:1371–1379. [[PubMed: 20502870](#)]
-
54. Veelo D, Dongelmans DA, Binnekade JM, et al. Adaptive support ventilation: a translational study evaluating the size of delivered tidal volume. *Int J Artif Organs*. 2010;33:302–309. [[PubMed: 20593352](#)]
-
55. Tassaux D, Dalmas E, Gratadour P, Jolliet P. Patient-ventilator interactions during partial ventilatory support: a preliminary study comparing the effects of adaptive support ventilation with synchronized intermittent mandatory ventilation plus inspiratory pressure support. *Crit Care Med*. 2002;30:801–807. [[PubMed: 11940749](#)]
-
56. Laubscher TP, Heinrichs W, Weiler N, et al. An adaptive lung ventilation controller. *IEEE Trans Biomed Eng*. 1995;41:51–59.
-
57. Laubscher TP, Frutiger A, Fanconi S, et al. Automatic selection of tidal volume, respiratory frequency and minute volume in intubated ICU patients as startup procedure for closed-loop controlled ventilation. *Int J Clin Monit Comput*. 1994;11:19–30. [[PubMed: 8195655](#)]
-
58. Laubscher TP, Frutiger A, Fanconi S, Brunner JX. The automatic selection of ventilation parameters during the initial phase of mechanical ventilation. *Intensive Care Med*. 1996;22:199–207. [[PubMed: 8727432](#)]
-
59. Weiler N, Eberle B, Latorre F, et al. [Adaptive lung ventilation (AVL). Evaluation of new closed loop regulated respiration algorithm for operation in the hyperextended lateral position] [In German.]. *Anaesthesist*. 1996;45:950–956. [[PubMed: 8992909](#)]
-
60. Weiler N, Eberle B, Heinrichs W. Adaptive lung ventilation (ALV) during anesthesia for pulmonary surgery: automatic response to transitions to and from one-lung ventilation. *Int J Clin Monit Comput*. 1998;14:245–252. [[PubMed: 9754613](#)]
-
61. Weiler N, Heinrichs W, Kebler W. The ALV mode: a safe closed loop algorithm for ventilation during total intravenous anesthesia. *Int J Clin Monit Comput*. 1994;11:85–88. [[PubMed: 7930854](#)]
-
62. Belliato M, Maggio G, Neri S, et al. Evaluation of the adaptive support ventilation (ASV) mode in paralyzed patients. *Intensive Care Med*. 2000;26:S327.
-
63. Linton DM, Brunner JX, Laubscher TP. Continuous use of an adaptive lung ventilation controller in critically ill patients in a multi-disciplinary intensive care unit. *S Afr Med J*. 1995;85:430–433.
-
64. Dongelmans DA, Paulus F, Veelo DP, et al. Adaptive support ventilation may deliver unwanted respiratory rate-tidal volume combinations in patients with acute lung injury ventilated according to an open lung strategy. *Anesthesiology*. 2011;114:1138–1143. [[PubMed: 21383619](#)]
-
65. Jaber S, Sebbane M, Verzilli D, et al. Adaptive support ventilation and pressure support ventilation behavior in response to increased ventilator demand. *Anesthesiology*. 2009;110:620–627. [[PubMed: 19225395](#)]
-

66. Dongelmans DA, Veelo DP, Binnekade JM, et al. Adaptive support ventilation with protocolized de-escalation and escalation does not accelerate tracheal extubation of patients after non fast track cardiothoracic surgery. *Anesth Analg*. 2010;111:961–967. [[PubMed: 20810675](#)]

67. Linton DM, Potgieter PD, Davis S, et al. Automatic weaning from mechanical ventilation using an adaptive lung controller. *Chest*. 1994;106:1843–1850. [[PubMed: 7988211](#)]

68. Sulzer CF, Chiolerio R, Cassot PG, et al. Adaptive support ventilation for fast tracheal extubation after cardiac surgery: a randomized, controlled study. *Anesthesiology*. 2001;95:1339–1345. [[PubMed: 11748389](#)]

69. Cassina T, Chiolerio R, Mauri R, Revelly JP. Clinical experience with adaptive support ventilation for fast-track cardiac surgery. *J Cardiothorac Vasc Anesth*. 2003;17:571–575. [[PubMed: 14579209](#)]

70. Petter AH, Chiolerio RL, Cassina T, et al. Automatic “respirator/weaning” with adaptive support ventilation: The effect on duration of endotracheal intubation and patient management. *Anesth Analg*. 2003;97:1743–1750. [[PubMed: 14633553](#)]

71. Bersten AD, Rutten AJ, Vedig AE, Skowronski GA. Additional work of breathing imposed by endotracheal tubes, breathing circuits, and intensive care ventilators. *Crit Care Med*. 1989;17:671–680. [[PubMed: 2736929](#)]

72. Shapiro M, Wilson RK, Casar G, et al. Work of breathing through different sized endotracheal tubes. *Crit Care Med*. 1986;14:1028–1031. [[PubMed: 3780244](#)]

73. Bersten AD, Rutten AJ, Vedig AE. Efficacy of pressure support in compensating for apparatus work. *Anaesth Intensive Care*. 1993;21:67–71. [[PubMed: 8447610](#)]

74. Brochard L, Rua F, Lorini H, et al. Inspiratory pressure support compensates for the additional work of breathing caused by the endotracheal tube. *Anesthesiology*. 1991;75:739–745. [[PubMed: 1952198](#)]

75. Fiastro JF, Habib MP, Quan SF. Pressure support compensation for inspiratory work due to endotracheal tubes and demand continuous positive airway pressure. *Chest*. 1988;93:499–505. [[PubMed: 3277803](#)]

76. Guttman, Eberhard L, Fabry B, et al. Continuous calculation of intratracheal pressure in tracheally intubated patients. *Anesthesiology*. 1993;79:503–513. [[PubMed: 8363076](#)]

77. Guttman J, Haberthur C, Mols G. Automatic tube compensation. *Respir Care Clin N Am*. 2001;7:475–483. [[PubMed: 11517035](#)]

78. Guttman J, Bernhard H, Mols G, et al. Respiratory comfort of automatic tube compensation and inspiratory pressure support in conscious humans. *Intensive Care Med*. 1997;23:1119–1123. [[PubMed: 9434916](#)]

79. Fabry B, Guttman J, Eberhard L, Wolff G. Automatic compensation of endotracheal tube resistance in spontaneously breathing patients. *Technol Health Care*. 1994;1:281–285.

80. Fabry B, Haberthur G, Zappe D, et al. Breathing pattern and additional work of breathing in spontaneously breathing patients with different ventilatory demands during inspiratory pressure support and automatic tube compensation.

Intensive Care Med. 1997;23:545–550. [PubMed: 9201527]

81. Stocker R, Fabry B, Eberhard L, Haberthur C. Support of spontaneous breathing in the intubated patient: automatic tube compensation (ATC) and proportional assist ventilation (PAV). *Acta Anaesthesiol Scand Suppl.* 1997;111:123–129. [PubMed: 9420983]

82. Takahashi T, Takezawa J, Kimura T, et al. Comparison of inspiratory work of breathing in T-piece breathing, PSV, and pleural pressure support ventilation (PPSV). *Chest.* 1991;100:1030–1034. [PubMed: 1914553]

83. Esquer C, Claire N, D'Ugard C, et al. Mechanisms of hypoxemia episodes in spontaneously breathing preterm infants after mechanical ventilation. *Neonatology.* 2008;94(2):100–104. [PubMed: 18277057]

84. Mitamura Y, Mikami T, Yamamoto K, Mimura K. A dual control system for assisting respiration. *Med Biol Eng.* 1975;13(6):846–854.

85. Beddis IR, Collins P, Levy NM, et al. A new technique for servo-control of arterial oxygen tension in preterm infants. *Arch Dis Child.* 1979;54:278–280. [PubMed: 453911]

86. Collins P, Levy NM, Beddis IR, et al. Apparatus for the servocontrol of arterial oxygen tension in preterm infants. *Med Biol Eng Comput.* 1979;17:449–452. [PubMed: 316056]

87. Sano A, Kikucki M. Adaptive control of arterial oxygen pressure of newborn infants under incubator oxygen treatments. *Proc IEEE.* 1985;132(Pt. D., No. 5):205–211.

88. Yu C, He WG, So JM, et al. Improvement in arterial oxygen control using multiple model adaptive control procedures. *IEEE Trans Biomed Eng.* 1987;34(8):567–574. [PubMed: 3623584]

89. Morozoff PE, Evans RW. Closed-loop control of SaO₂ in the neonate. *Biomed Instrum Technol.* 1992;26:117–123. [PubMed: 1562825]

90. Dugdale RE, Cameron RG, Lealman GT. Closed loop control of the partial pressure of arterial oxygen in neonates. *Clin Phys Physiol Meas.* 1988;9:291–305. [PubMed: 3233884]

91. Raemer DB, Ji X, Topulos GP. Flx controller: An instrument to automatically adjust inspired oxygen fraction using feedback control from a pulse oximeter. *J Clin Monit.* 1997;13:91–101. [PubMed: 9112204]

92. Claire N, Gerhardt T, Everett R, et al. Closed-loop controlled inspired oxygen concentration for mechanically ventilated very low birth weight infants with frequent episodes of hypoxemia. *Pediatrics.* 2001;107(5):1120–1124. [PubMed: 11331696]

93. Claire N, D'Ugard C, Bancalari E. Automated adjustment of inspired oxygen in preterm infants with frequent fluctuations in oxygenation: a pilot clinical trial. *J Pediatr.* 2009;155(5):640–645. [PubMed: 19595375]

94. Claire N, Bancalari E, D'Ugard C, et al. Multicenter crossover study of automated control of inspired oxygen in ventilated preterm infants. *Pediatrics.* 2011;127(1):e76–e83.

95. Claire N. Automated regulation of inspired [oxygen](#) in preterm infants: oxygenation stability and clinician workload. *Anesth Analg*. 2007;105:s37–s41.
-
96. Johannigman JA, Branson RD, Beck G, Lacroy D. [Oxygen](#) conservation with autonomous control of inspired [oxygen](#) concentration in ventilated trauma patients. *J Trauma*. 2009;66:386–392. [[PubMed: 19204511](#)]
-
97. Johannigman JA, Branson RD, Edwards MG. Closed loop control of inspired [oxygen](#) concentration in trauma patients. *J Am Coll Surg*. 2009;208(5):763–768. [[PubMed: 19476833](#)]
-
98. de Graaf AE, Dongelmans DA, Binnekade JM, de Jonge E. Clinicians response to hyperoxia in ventilated patients in a Dutch ICU depends on the level of FI_{O_2} . *Intensive Care Med*. 2011;37:43–47.
-
99. Branson RD, Robinson BR. Oxygen—when is more the enemy of good? *Intensive Care Med*. 2011;37(1):1–3. [[PubMed: 20878145](#)]
-
100. Lellouche F, Hernert P, Jouvett P, et al. Clinical evaluation of a computer-driven algorithm to automatically set PEEP and FI_{O_2} during mechanical ventilation. *Am J Respir Crit Care Med*. 2009;179:A3652.
-
101. Younes M. Proportional assist ventilation, a new approach to ventilatory support. *Am Rev Respir Dis*. 1992;145:114–120. [[PubMed: 1731573](#)]
-
102. Sinderby C. Neurally adjusted ventilatory assist (NAVA). *Minerva Anesthesiol*. 2002;68:378–380. [[PubMed: 12029249](#)]
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